

Week 10 Lecture Plan

Network = Concurrency + Failure: Sockets, Latency, and Toxiproxy

Instructor: Dr. Mohamad Aoude

Pre-Class Preparation

- ☐ Week 10 lab compiles: `week10-lab/week10-day1-network-concurrency-failure-toxiproxy`
- ☐ Smoke test passes: `gradlew.bat test`
- ☐ Instructor check passes: `gradlew.bat -Pinstructor=true instructorCheck`
- ☐ Docker available if showing Toxiproxy live.
- ☐ Lecture PDFs and lab manual ready.

Learning Targets

1. State the three laws of networks.
2. Explain why a socket boundary is not shared memory.
3. Convert local queue semantics into a socket protocol.
4. Use Toxiproxy as a controlled failure-injection tool.
5. Measure throughput, latency, and timeout rate under injected failure.

Lecture Structure

Segment	Material	Goal
Part 1	Lecture_10_01_nar	Network laws and phase transition
Part 2	Lecture_10_02_nar	Queue-to-socket protocol and loopback demo
Part 3	Lecture_10_03_nar	Toxiproxy latency, bandwidth, reset, timeout
Part 4	Lecture_10_04_nar	Measurement summary and Week 11 bridge

Exit Ticket

1. Which network law broke your first design assumption?
2. What does the socket protocol make explicit that a local queue hides?
3. Which metric changed most under latency injection?